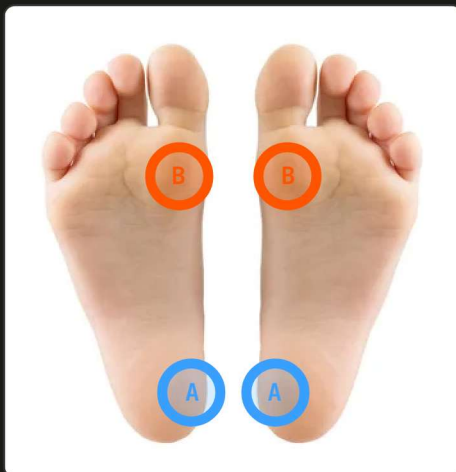


# THE PERFECT SQUAT

MAXIMUM QUAD TENSION – WITHOUT KNEE OR BACK PAIN



## 01 FOOT PRESSURE – YOUR FOUNDATION



A INNER HEEL · B BIG-TOE BALL

Your foot contact decides whether the tension lands in the **quadriceps**. The goal is a slight, spiral **internal rotation (IR)** – created by this rolling sequence as you descend:

### A START: INNER HEEL

At the start of the descent, the weight sits on the inner heel.

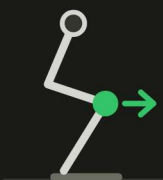
### B TARGET: BIG-TOE BALL

The deeper you go, the more the pressure shifts onto the ball of the big toe (first metatarsal head). At the bottom, the arch is allowed to flatten slightly – that's intended.

*If you push the knees/feet outward or shift weight to the outer edge, you force external rotation – the tension moves to the adductors & glutes. But: the little toe always keeps light ground contact – the foot must never collapse fully inward.*

## 02 KNEES FORWARD – AND KEEP THEM THERE

Yes, the knees may travel past the toes. In fact they must – continuously, throughout the entire descent.



### ✓ CORRECT

The knees travel **steadily forward over the toes**. As long as they stay forward, the mechanical tension stays in the quad.



### ✗ WRONG

You go deeper, but the knees **suddenly shift back** → tension leaves the quad, load moves to the lower back & hips.

## 03 HOW DEEP SHOULD YOU SQUAT?

"Ass-to-grass" at any cost? No. The ECM says: **as deep as your biomechanics cleanly allow** — not one centimeter deeper.



### CLEAN DEPTH

PELVIS MOVES "WITHIN ITSELF"

A perfect, deep squat allows **"relative motion"** in the pelvis: at the lowest point the sacrum tilts backward (counternutation) while the ilia rotate internally/externally around it.



### THE BUTT WINK

PELVIS ROLLS AWAY AS A BLOCK

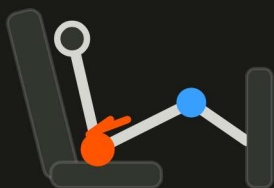
If there's no room for this internal rotation, the pelvis no longer moves within itself but rolls as a **rigid block backward under the body**. The quadriceps loses its tension — your joints and axial skeleton have to compensate for the load.

**90-100°**  
HIP FLEXION

### YOUR STOP RULE

Focus on the sole of your foot: as soon as **the knees stop traveling forward and the heels want to lift**, you've reached the end of your effective range of motion — usually around 90-100° of hip flexion. Every centimeter deeper tucks the pelvis and sends the tension into the lumbar spine.

## 04 LEG PRESS & HACK SQUAT: COUNTERFORCE



PELVIS INTO THE PAD

- **Place feet low & straight:** Rather low on the platform (greater lever for the quads); feet and knees point *relatively* straight forward.
- **Keep your butt down.** Lower the weight only as far as the pelvis stays firmly on the pad — never let it lift to force artificial extra depth.
- **Actively press back:** The lower back and pelvis are deliberately pressed firmly into the pad the whole time — that's the counterforce.
- **The result:** Back locked + knees forward = isolated compression in the quadriceps — without shear forces on the spine.

## 05 YOUR BUILD HAS A SAY

Who can squat deep and upright isn't a question of discipline — it's a question of **bone structure**. So don't compare yourself to others; work with your build.

### NARROW RIB CAGE

NARROW ISA • "FORCE ABSORBER"

The natural squatter: the pelvis starts in a favorable position with **plenty of room for internal rotation**. Can often squat extremely upright and very deep — without a butt wink.

### WIDE RIB CAGE

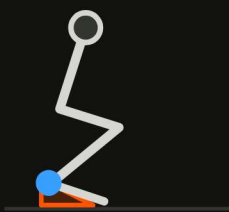
WIDE ISA • "FORCE PRODUCER"

Naturally tilted forward (anterior pelvic tilt) — the pelvis **lacks the room to rotate**. Deep squats are hard; hinge movements like the deadlift suit you better.

**What is the ISA?** The infrasternal angle — the angle of your lower rib arch.

**Narrow:** The pelvis starts slightly tilted back (counternutation) — lots of room for internal rotation; deep squats come naturally to you.

**Wide:** The pelvis starts tilted forward (anterior tilt) — little rotation room, strong in the hinge (deadlift); for squatting, wedges are almost mandatory for you.



### THE TRICK: WEDGES (HEEL WEDGES)

A wedge or a raise under the heels shifts your center of gravity backward and brings the pelvis into a **more neutral starting position**. Result: more room to sink cleanly deep — without a butt wink. Especially for "wide" types, a game changer.

## 06 BREATHING: INHALE DOWN, EXHALE UP

### DESCENT: INHALE



Deep through the nose. The back of the rib cage fills (expansion), the spine stays upright — the pelvis is optimally prepared.

### ASCENT: EXHALE



Exhale as you press up — this supports the natural compression and delivers the force upward.

# 07 THE QUAD SWEEP

The "sweep" — the visible outer quadriceps (**vastus lateralis**) — isn't built by pushing the knees forward alone, but through the full rotational chain.

## THE MECHANISM

AXIAL INTERNAL-ROTATION SPIRAL

Pelvis (ilium), thigh (femur), shin (tibia) and foot rotate **inward together** while the arch flattens (pronation). This anterior compression aligns the bones so the quadriceps has to take up the entire mechanical load.

## THE MOST COMMON MISTAKE

TURNING OUT FEELS STABLE

Turning the feet or knees far outward feels more stable — but the anterior compression is lost. The force shifts backward onto the **adductors & glutes** — the quad stays flat.

## KEY CUE FOR THE SWEEP

Knees forward **and** let the whole chain turn slightly inward — inner heel, big-toe ball, flattening arch. It's not the wide stance that builds the outer quadriceps, but internal rotation under load.

## SUMMARY — YOUR SQUAT CHECKLIST

### FOOT

Inner heel → big-toe ball. The arch may flatten slightly at the bottom.

### KNEES

Continuously forward over the toes — never let them retreat.

### DEPTH

Stop before the pelvis rolls away — usually around 90–100° of hip flexion.

### MACHINE

Press the pelvis firmly into the pad — never buy depth by lifting off.

### STACK & GAZE

Rib cage down, no arched back — look to the horizon, not the floor.

### TEMPO

Slow down, controlled up — no bounce out of the lower back.